

Smart Attendance System (React.js + Supabase)

1) Project Overview

This project delivers a modern attendance platform that allows students to mark attendance by **scanning a time-rotating QR/Barcode displayed on the teacher's dashboard**. The code refreshes every **5 10 seconds** to prevent screenshots or sharing. The system also includes a **presence verification mechanism** to ensure only students physically in class can mark attendance.

The platform includes:

- **Teacher Dashboard:** create/manage classes, run live attendance sessions, and view analytics.
- **Student Dashboard:** view attendance history by class and teacher.
- **Supabase Database:** secure data storage, authentication, and row-level security.

2) Key Objectives

- Make attendance **fast and effortless** (scan → submit → success).
- Prevent false attendance through:
 - **Rotating QR/Barcode**
 - **Session-based tokens**
 - **Location / proximity verification**
 - **Rate limiting + duplicate-scan prevention**
- Support **multiple teachers and multiple students**, with correct relationship mapping.
- Provide clear dashboards for **tracking, reporting, and exporting**.

3) Core Features

A) Teacher Dashboard

Class & Scheduling Management

- Create/edit classes (Class Title, Section, Subject, etc.)
- Set class schedule (days/dates/times)
- View class roster (students assigned)

Live Attendance Session Mode

- “Start Session” button generates a **Live QR/Barcode**

- QR/Barcode **changes every 5–10 seconds**
- Session controls:
 - Start / Pause / End
 - Session duration limit (e.g., 10–20 minutes)
 - Late marking rules (optional)

Attendance Monitoring

- Live list of students:
 - Present / Absent / Late (optional)
 - Student Name, Student ID, Timestamp
- Quick summary counters:
 - Total students
 - Present count
 - Absent count
- Search/filter by student name/ID

Attendance Records & Reports

- View attendance by:
 - Date
 - Class
 - Student
- Export options (CSV/PDF) (optional)
- Teacher can see historical trends (optional)

B) Student Dashboard

Attendance Marking

- Scan teacher's live rotating QR/Barcode using phone camera
- Automatic attendance submission once verified
- Confirmation message + timestamp

Attendance History & Details

- View attendance records by:
 - Class name/title
 - Teacher name
 - Date and status
- Summary stats:
 - Total attended classes
 - Attendance percentage per class (optional)

C) Multi-Teacher / Multi-Student Mapping

The system supports:

- Many teachers → many classes
- Many students → many classes
- A student can be enrolled in multiple teachers' classes
- A teacher can teach multiple classes and have many students

This is achieved via a **Class** entity and **Enrollment (join table)** in the database.

4) Anti-Fraud & Validation Logic

A) Rotating QR/Barcode (Every 5–10 seconds)

- Teacher dashboard displays a QR code containing a **short-lived token** (expires fast).
- Token rotates frequently and is valid only for a specific class session.
- Screenshot sharing becomes useless because tokens expire quickly.

B) Session-Based Token Verification

When student scans:

1. Student app reads token.
2. App sends token + student ID + class/session ID to backend.
3. Backend verifies:
 - Token is valid
 - Token is not expired
 - Token belongs to the teacher's currently running session
 - Student is enrolled in that class
 - Student has not already been marked present for that session

C) “Student is physically present” verification options

You can implement one or combine multiple methods depending on budget and device constraints:

Option 1: GPS Geofencing (Recommended baseline)

- Teacher device captures classroom location when session starts.
- Student device must be within an allowed radius (e.g., 15–40 meters).
- Pros: Simple, scalable
- Note: GPS indoors can vary, so we allow a reasonable radius.

Option 2: Network Verification (Wi-Fi)

- If campus Wi-Fi exists, verify student is connected to the same Wi-Fi SSID/BSSID range (where available).
- Pros: Strong in indoor environments
- Cons: Device/platform limitations sometimes restrict exact network details.

5) Technology Stack

Frontend

- **React.js** (Teacher + Student dashboards)
- QR generation: `qrcode.react` or similar
- QR scanning on student side: `html5-qrcode` / `react-qrcode`
- UI: Material UI / Tailwind CSS (based on preference)

Backend / Database

- **Supabase**
 - Authentication (Teachers/Students login)
 - Postgres database
 - Row Level Security (RLS)
 - Edge Functions (optional, for secure token generation)

Security

- Short-lived rotating tokens
- Enforced server-side validation
- Role-based access (teacher vs student)
- RLS policies to protect attendance records

6) Suggested Database Design (Supabase)

Core tables:

- `users` (linked to Supabase auth)
- `teachers` (profile info)
- `students` (profile info)
- `classes` (title, `teacher_id`, schedule metadata)
- `enrollments` (`class_id`, `student_id`) ← many-to-many mapping
- `sessions` (`class_id`, `teacher_id`, `start_time`, `end_time`, `location_lat/lng`, status)
- `session_tokens` (`session_id`, `token_hash`, `expires_at`) ← rotating tokens
- `attendance` (`session_id`, `student_id`, status, `marked_at`, `verification_method`, `device_info` optional)

7) User Flow Summary

Teacher Flow

1. Teacher logs in → selects class → clicks **Start Attendance Session**
2. System begins generating **rotating QR/Barcode**
3. Teacher watches live updates as students mark attendance
4. Teacher ends session → attendance is locked and stored

Student Flow

1. Student logs in → opens “Mark Attendance”
2. Scans QR → system checks:
 - token validity
 - enrollment
 - proximity verification
 - duplicate marking
3. Attendance confirmed and visible in student history

8) Project Delivery Plan (Phases)

Phase 1: Planning & UI/DB Setup

- Finalize dashboards layout
- Build Supabase schema + roles
- Authentication & access control

Phase 2: Teacher Dashboard (Core)

- Class creation + scheduling
- Roster/enrollment view
- Session start/end UI

Phase 3: Student Attendance Scanning

- QR scanning
- Attendance submit + confirmations
- Student attendance history dashboard

Phase 4: Anti-Fraud Verification

- Token rotation logic (5–10 seconds)
- Proximity verification implementation (GPS / optional PIN)
- Duplicate prevention + rate limiting

Phase 5: Reports & Polish

- Teacher analytics (present/absent counts, filters)
- Export (optional)
- QA testing + deployment

THE END